



Reaction kinetic model of glycolysis of polyethylene terephthalate

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ABSTRACT

The reaction kinetics of the glycolysis of polyethylene terephthalate (PET) in ethylene glycol (EG) solvent using zinc acetate as a catalyst were investigated. The experimental results show that stirring rate, EG/PET mass ratio, and catalyst dosage all influence the glycolysis rate. The reaction temperature has a decisive effect on the glycolysis rate and the yield of the target product, bis(2-hydroxyethyl) terephthalate (BHET). By tracking the distribution of PET repeat units in the products and combining the heterogeneous shrinking core model and homogeneous reaction model, a reaction kinetic model for PET glycolysis was developed. The model accurately predicts the equilibrium product distribution at high temperature and the reaction behavior of large PET particles. The activation energy for the depolymerization of PET to oligomers is 172.71 kJ/mol, making it the rate-determining step of the entire reaction network. The standard reaction enthalpy and entropy change for this step are 22.58 kJ/mol and 16.46 J/mol·K, respectively, indicating that PET glycolysis is an entropy-driven process that requires high temperatures. Increasing the reaction temperature significantly reduces the sensitivity of the reaction rate to PET particle diameter. When conducting PET glycolysis in a series of continuous stirred tank reactors, it is necessary to use both high temperatures and small PET particle sizes to minimize reactor volume and reduce the number of reactors in series.

1. Introduction

Polyethylene terephthalate (PET) is one of the most widely used thermoplastic polymers. It is primarily employed in the production of beverage and food containers, due to its excellent barrier properties and recyclability, making it the preferred material for bottles, containers, and films (Siracusa et al., 2008; Mouarbes et al., 2019; Geyer et al., 2017). In the textile industry, PET is crucial for manufacturing synthetic fibers such as polyester, underlining its versatility and economic significance (Tsironi et al., 2022; Mougenot et al., 1996; Mulford and Chen, 2011). Currently, global PET production exceeds 70 million tons annually, and this figure is projected to rise as industrial applications and consumer demand grow. However, the widespread use of PET presents significant environmental challenges (Javed et al., 2023). The resistance to biodegradation of PET means that discarded products persist in the

environment, breaking down into microplastics that infiltrate water sources, soil, and the food chain, posing threats to ecosystems and wildlife. For instance, microplastics can be ingested by marine organisms, leading to bioaccumulation and adverse health effects (Suhaimi et al., 2022; Mason et al., 2018; Martín et al., 2021; Qi et al., 2020).

Recycling PET is essential for reducing plastic waste and conserving resources, mainly relying on physical (mechanical) and chemical methods (Raheem et al., 2019; Paszun and Szychaj, 1997). Physical recycling involves collecting, sorting, cleaning, and reprocessing PET waste into secondary products without altering its chemical structure. However, recycled PET can undergo polymer chain degradation due to thermal and mechanical stress, resulting in reduced molecular weight, decreased mechanical properties, and impaired transparency. This limits its use in high-quality applications such as food-grade packaging. Chemical recycling, including hydrolysis, glycolysis, methanolysis, and

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pyrolysis, offers alternative solutions (George and Kurian, 2014; Uekert et al., 2023). Glycolysis, in particular, has gained significant attention due to its mild operating conditions, ease of product separation and purification, and process flexibility (Xi et al., 2005; McNeeley and Liu, 2024; Kim et al., 2022).

Glycolysis of PET typically employs an excess of ethylene glycol (EG) as both a solvent and depolymerizing agent. Catalysts include metal acetates, metal oxides, salts, and organic catalysts such as ionic liquids (López-Fonseca et al., 2010; Xin et al., 2021; Le et al., 2022). The reaction is conducted at temperatures of 180 to 240 °C and pressures of 0.1 to 0.3 MPa, depending on catalyst efficiency, PET particle size, and the target product. The primary product of PET glycolysis is bis (2-hydroxyethyl) terephthalate (BHET), which can be further purified for use as a precursor in PET production. Detailed studies have explored the reaction mechanism of zinc acetate as a representative catalyst in PET glycolysis (Imran et al., 2013; Clark and Shaver, 2024). The process initiates with Zn^{2+} ions coordinating to the carbonyl oxygen atoms of ester bonds in PET. This interaction weakens the carbon–oxygen bond within the ester, making it more susceptible to nucleophilic attack by EG. Acting as a nucleophile, EG targets the carbonyl carbon in the PET ester bond, initiating the transesterification process. Ethylene glycol replaces the ester bond, breaking down the PET chain to form the primary product, BHET (George and Kurian, 2014; Kim et al., 2022; Al-Sabagh et al., 2014).

Compared to the extensive research on catalytic reaction mechanisms, there are relatively few studies on the reaction kinetics of PET degradation. Notable studies using the pseudo-homogeneous model to investigate PET degradation include reports by Goje, López-Fonseca, Wan, and Held (López-Fonseca et al., 2010; Goje and Mishra, 2003; López-Fonseca et al., 2011; Wan et al., 2001; Mishra and Goje, 2003; Schlüter et al., 2024; Schlüter et al., 2024). Goje et al. explored the batch-mode glycolysis of PET at atmospheric pressure and temperatures ranging from 100 to 220 °C, optimizing catalyst types. They found that the depolymerization rate of PET decreased with increasing particle size and was influenced by the concentration of EG. The depolymerization activation energy, based on the pseudo-homogeneous model, was determined to be 46.2 kJ/mol (Goje and Mishra, 2003; Mishra and Goje, 2003). López-Fonseca examined the glycolysis of PET at temperatures between 165 and 196 °C, using sodium carbonate as a catalyst. Their analysis, based on a pseudo-homogeneous power law model, indicated an activation energy of 185 kJ/mol for the forward depolymerization reaction. Furthermore, the glycolysis process was endothermic, with a standard reaction enthalpy of 12 kJ/mol (López-Fonseca et al., 2010; López-Fonseca et al., 2011). Wan et al. studied the depolymerization of PET sheets in potassium hydroxide solution at temperatures of 120 to 160 °C and pressures of 1.7 to 4.6 atm. The experiments revealed that the residual solids retained their original morphology, and their molecular weight was close to the original PET value. The study proposed that the primary reaction occurred at the polymer chain ends on the surface of solid PET. Reaction kinetics analysis using the pseudo-homogeneous model showed that the depolymerization rate was first-order with respect to the concentrations of potassium hydroxide and PET, with an activation energy of 69 kJ/mol (Wan et al., 2001). Held et al. used a co-solvent γ -valerolactone in the PET glycolysis, finding that the reaction kinetics were boosted by almost two orders of magnitude compared to the standard literature process. The ePC-SAFT equation of state was employed to develop an activity-based model for PET glycolysis. The thermodynamic activity of the catalyst zinc acetate was included, enabling the incorporation of catalyst interactions in the liquid reaction phase (Schlüter et al., 2024; Schlüter et al., 2024).

Some studies introduced heterogeneous models to examine PET degradation (Genta et al., 2010; Yoshioka et al., 2001). Yoshioka et al. investigated the degradation of PET powder in 7 to 13 M nitric acid at 70 to 100 °C under atmospheric pressure, using a chemical reaction-controlled shrinking core model to explain hydrolysis behavior (Yoshioka et al., 1998). They found that the effective surface area was

proportional to the yield of unreacted PET, and the hydrolysis rate constant was inversely proportional to PET particle size and nitric acid concentration, with an activation energy of 101.3 kJ/mol. Ruvo-Filho examined the NaOH-catalyzed glycolysis of PET at 150 to 185 °C, concluding that the rate-determining step was the chemical reaction. The depolymerization process was described using a shrinking core model, which accounted for the formation and growth of cracks and pores on the polymer surface. The reaction kinetics analysis indicated an activation energy of 172.7 kJ/mol, with thermodynamic analysis showing that entropy changes drove the reaction to overcome the high potential energy barrier (Ruvo-Filho and Curti, 2006). Viana et al. studied the glycolysis of PET using zinc acetate as a catalyst. The study classified particles into different size ranges and proposed a relationship between particle surface area, conversion rate, and depolymerization rate. They suggested that a reaction equilibrium existed between oligomers and unreacted PET, leading to a decreased conversion rate. The proposed heterogeneous model distinguished different types of active sites on the PET surface (Viana et al., 2011).

The melting point of PET ranges from 250 to 255 °C, and PET remains essentially insoluble in EG below 230 °C (Schlüter et al., 2024; Campanelli et al., 1994). Consequently, the glycolysis of PET in EG initially proceeds as a liquid–solid two-phase reaction. The molecular weight distribution of unreacted PET closely resembles that of the raw material, suggesting that the liquid–solid reaction follows the characteristics of the shrinking core model (Wan et al., 2001). Furthermore, the presence of oligomers in the liquid phase indicates that homogeneous reactions also play a significant role in PET glycolysis (Viana et al., 2011). These observations demonstrate that neither a pseudo-homogeneous reaction model nor a heterogeneous shrinking core model alone can fully capture the glycolysis behavior of PET.

Kinetic modeling of glycolysis is fundamental for process optimization and reactor design (McNeeley and Liu, 2024). However, existing studies often lack a direct connection between kinetic models and their practical engineering applications. While most studies focus on PET depolymerization at lower temperatures, industrial applications typically operate at higher temperatures above 200 °C to improve reaction efficiency. Thus, it is essential to investigate the prediction and optimization of depolymerization behavior at high temperatures based on reaction kinetic models.

To address the above challenges, the reaction kinetics of PET glycolysis in EG solvent using zinc acetate as a catalyst were investigated. Initially, the effects of stirring rate, PET/EG ratio, and catalyst dosage on PET degradation were evaluated. The glycolysis behavior of PET was then examined within the temperature range of 140 to 180 °C, with the presence of BHET and oligomers in the products confirmed through physical characterization. A reaction kinetic model for PET glycolysis was developed by coupling the heterogeneous shrinking core model with the homogeneous reaction model. After that, model parameters were regressed and the reliability of the model was assessed to ensure accuracy. Based on the derived reaction kinetic parameters, further recommendations for the optimal process conditions for glycolysis in both batch and multi-reactor series continuous stirred tank reactor (CSTR) modes were proposed.

2. Material and methods

The experiments used two types of spherical PET particles, with average diameters of 0.15 and 0.85 mm, purchased from Suzhou Yuan Plastic Co., Ltd., as representatives of PET. The PET had a molecular weight of 25,000 Da and a density of 1.37 g/cm³. EG and zinc acetate were purchased from Shanghai Macklin Biochemical Co., Ltd. The glycolysis reactions were conducted in batch mode using a 100 mL Parr 4598 reactor. This reactor was equipped with temperature and pressure sensors, along with a four-blade flat paddle stirrer. A heating jacket of 0.7 kW efficiently heated the reaction system from room temperature to 200 °C in 15 min. To establish the reaction kinetic model, experiments

were carried out at temperatures of 140, 160, and 180 °C.

The typical procedure for PET glycolysis is as follows. Firstly, add 10 g of PET, 40 g of EG, and 0.1 g of zinc acetate into the reactor. After purging the reaction system with high-purity nitrogen to remove oxygen, seal the reactor and begin preheating. Once the reaction system reaches the preset reaction temperature, start timing the reaction. After the specified reaction time, quench the reaction by rapidly cooling the reactor body with water. The typical product separation process is as follows. Firstly, the product was preheated to 95 °C for the first filtration, where unreacted PET was obtained on filter paper with a pore size of 20 µm, and filtrate I is collected. Filtrate I was then heated to 80 °C for the second filtration, where oligomers were obtained on the filter paper, resulting in filtrate II. Filtrate II was left to stand at 0 °C for 12 h before undergoing a third filtration, where the product BHET was obtained on the filter paper.

Based on material balance, experiments with a mass balance of $\geq 98\%$ are considered valid. Experiments under the same conditions were repeated three times to evaluate the impact of experimental error. The conversion rate of PET (X_{PET}) and the yield of BHET (P_{BHET}) are calculated by

$$X_{\text{PET}} = \frac{m_{\text{PET,ini}} - m_{\text{PET}}}{m_{\text{PET,ini}}} \times 100\% \quad (1)$$

$$P_{\text{BHET}} = \frac{m_{\text{BHET}}/M_{\text{BHET}}}{m_{\text{PET,ini}}/M_{\text{PET}}} \times 100\% \quad (2)$$

where $m_{\text{PET,ini}}$ is the initial mass of PET, m_{PET} is the mass of PET obtained after the first filtration, and m_{BHET} is the mass of BHET obtained after the third filtration. M_{PET} and M_{BHET} are the molar masses of the repeat units of PET and BHET, respectively.

The solid product obtained from the third filtration was characterized using a nuclear magnetic resonance spectrometer (^1H NMR, Bruker Ascend 600, Germany) to confirm that the solid was the target product, BHET. The product obtained from the second filtration was characterized using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI TOF-MS, 4800 plus, ABS, Singapore) to confirm that the solid was oligomers and to determine the degree of polymerization.

In developing the reaction kinetic model for PET glycolysis, it is crucial to quantify the impact of the formation and consumption of oligomers, BHET, and EG on the overall liquid volume during depolymerization. This quantification allows for the calculation of the concentrations of these species in the liquid phase, which was achieved through quantum chemistry calculations. Firstly, the molecular structures of EG, BHET, and oligomers were optimized using the combination of the B3LYP functional and the 6-311 + G (d, p) basis set provided by Gaussian software. Subsequently, Multiwfn software was used to obtain the van der Waals volumes of each substance in the condensed phase at an electron density isosurface of 0.002 a.u. (Lu, 2024; Lu and Chen, 2012). These van der Waals volumes were then corrected based on experimental average density values of EG within the glycolysis reaction temperature range. The same correction factor was approximately applied to calculate the molecular molar volumes of BHET and oligomer.

3. Results and discussion

3.1. Performance of PET glycolysis under optimized conditions

Glycolysis of PET involves both heterogeneous and homogeneous reactions, with the apparent reaction rate influenced by factors such as liquid–solid mass transfer, surface chemical reactions, raw material ratios, and catalyst dosage. To determine optimal operating conditions and raw material ratios, preliminary experiments were conducted to investigate the effects of stirring rate, EG/PET mass ratio, and catalyst dosage on the glycolysis process. These experiments were performed at a

reaction temperature of 180 °C, a reaction time of 30 min, and a PET particle diameter of 0.15 mm. The experimental results, as shown in Fig. 1, provide insights into the influence of these parameters on the efficiency of the glycolysis process.

At lower reaction temperatures, PET glycolysis is primarily controlled by surface reaction kinetics (Yoshioka et al., 1998). However, as the reaction temperature rises, it is crucial to exclude the influence of liquid–solid mass transfer. Therefore, the effect of stirring rate on the PET degradation rate was investigated over a range of 50 to 300 rpm at an EG/PET mass ratio of 4/1 and a catalyst dosage of 1 wt% of the PET. As the stirring speed increases from 50 to 150 rpm, both the PET conversion rate and BHET yield show a slight increase. When the stirring speed is further increased to 200 rpm or higher, the PET conversion rate and BHET yield stabilize at approximately 88.5 % and 71.0 %, respectively. Accordingly, a stirring rate of 200 rpm may effectively eliminate the influence of liquid–solid mass transfer on the reaction kinetics, ensuring accurate kinetic modeling.

An appropriate EG/PET ratio helps increase the glycolysis rate, but an excessively high EG/PET ratio can escalate separation and recycling costs. To determine the optimal raw material ratio, PET glycolysis was conducted with EG/PET ratios ranging from 1:1 to 6:1 at a stirring rate of 200 rpm and catalyst dosage of 1 wt% of the PET. As the EG/PET ratio increases from 2:1 to 4:1, the PET conversion rate and BHET yield exhibit a steady upward trend. Notably, the BHET yield is significantly improved by approximately 15 %. However, when the EG ratio continues to increase, the change in PET conversion rate became negligible, while the BHET yield slightly declines.

Introducing a catalyst can accelerate the glycolysis process, but an excess amount may lead to product contamination. PET glycolysis with catalyst dosages ranging from 0 % to 1.2 % was applied at a stirring rate of 200 rpm and an EG/PET ratio of 4:1. In the absence of the catalyst, both the PET conversion rate and BHET yield are nearly zero. As the catalyst dosage increases, both metrics show a steady, monotonic rise. When the catalyst dosage reaches 1 wt% or higher of PET loading, the PET conversion rate and BHET yield no longer show significant changes.

Based on preliminary experimental results, the glycolysis kinetics of PET were investigated under the conditions of a stirring rate of 200 rpm, an EG/PET mass ratio of 4:1, and a catalyst dosage of 1 wt%. PET particles with a diameter of 0.15 mm were used for the experiments. The PET conversion rate and BHET yield at reaction temperatures of 140, 160, and 180 °C are shown in Fig. 2. The rate of PET glycolysis varies significantly with temperature. At a reaction temperature of 140 °C, PET conversion continues as the reaction time extends from 30 min to 240 min. When the reaction is terminated at 240 min, the PET conversion rate and BHET yield are 45.4 % and 35.8 %, respectively. Increasing the reaction temperature effectively accelerates the glycolysis process. At 180 °C, PET conversion and BHET yield reach 65.2 % and 51.6 % within just 5 min. Extending the reaction time to 40 min results in a PET conversion rate of 96.0 % and a BHET yield of 75.8 %. Apparently, further increasing the reaction temperature could introduce significant experimental errors in kinetic studies, as PET conversion occurs rapidly during the heating process, making precise measurements challenging.

3.2. Characterization on glycolysis products

To confirm that the solid obtained from the third filtration in the product separation process is BHET, it was characterized using ^1H NMR spectroscopy. The resulting spectrum is shown in Fig. 3.

In the ^1H NMR analysis, deuterated chloroform was used as the solvent. Its blank sample shows characteristic chemical shifts (δ) at 7.26, 1.59, and 0.00 ppm. On the ^1H NMR spectrum of the solid product, the characteristic peaks corresponding to the H atoms contained in the BHET molecule can be clearly observed. The peak with δ of 8.13 ppm corresponds to the H atoms on the benzene ring, the peaks with δ of 4.50 and 3.99 ppm correspond to the H atoms on the methylene group, and the peak with δ of 2.02 ppm corresponds to the H atoms on the hydroxyl

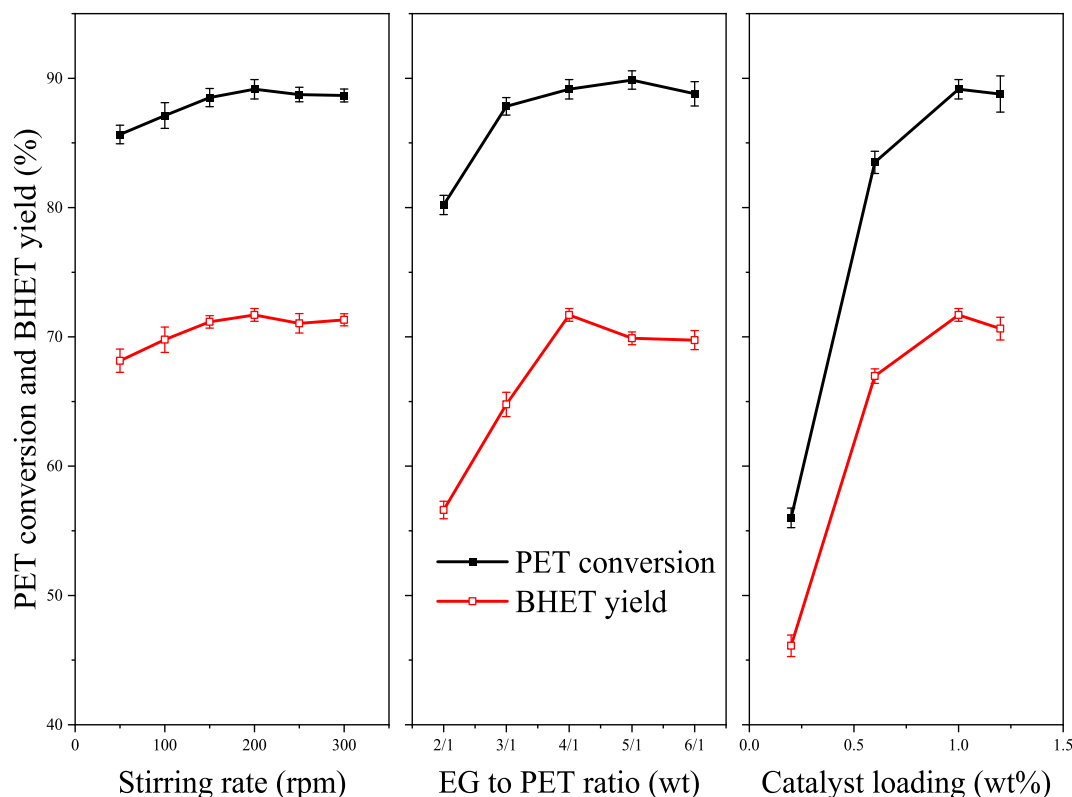


Fig. 1. PET conversion and BHET yield at different stirring rates (left), EG/PET ratios (middle) and catalyst dosages (right).

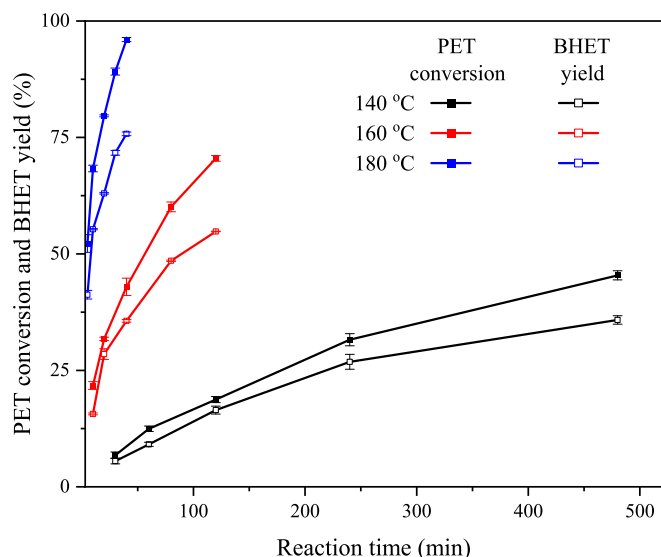


Fig. 2. PET conversion and BHET yield at different reaction temperatures.

group. The positions of the above characteristic peaks are in agreement with the literature values, so it can be confirmed that the obtained product is BHET (Wang et al., 2009). Since no other impurity peaks were observed, the separation method used in the experiment effectively yielded high-purity BHET.

The solid product obtained from the second filtration was characterized by MALDI TOF-MS, as shown in Fig. 4. During the analysis, an excess of sodium-containing matrix was added to the sample, causing the detected molecules to ionize into positively and negatively charged ions under laser irradiation. Consequently, the mass-to-charge ratio of each characteristic peak increases by the mass of a sodium atom.

As shown in the figure, distinct peaks corresponding to dimers, trimers, and tetramers are observed at mass-to-charge ratios of 469.237, 661.151, and 853.215, respectively (Pardal and Tersac, 2006). The difference between each peak is 192, corresponding to the molar mass of one PET repeat unit. At higher mass-to-charge ratios, trace amounts of higher-order oligomers can be identified. It can be confirmed that the products of PET glycolysis contain oligomers with a range of degrees of polymerization, predominantly in the form of dimers. Oligomers larger than tetramers are almost negligible.

3.3. Reaction kinetic model for PET glycolysis

3.3.1. Model development for PET glycolysis

The products of PET glycolysis can be categorized into unreacted PET particles, oligomers, and BHET. Unreacted PET retains its granular form throughout the reaction, with its diameter continually decreasing as the reaction progresses. Given sufficient reaction time, PET particles can be completely converted. Simultaneously, the concentrations of oligomers and BHET cease to change significantly with extended reaction time, indicating the existence of an equilibrium constraint between the two. Based on these observations, the glycolysis mechanism of PET can be proposed as consisting of heterogeneous shrinking core reaction steps and homogeneous reaction steps, as shown in Fig. 5.

At the beginning of the reaction, PET particles are dispersed in a solution composed of EG. After Zn^{2+} activating the carbonyl oxygen in the PET molecules on the surface of the particles, the subsequent random attack of the EG molecules will generate oligomers with a degree of polymerization of N . These oligomers, which present a certain distribution, then desorb from the particle surface into the liquid phase. In the reverse process of depolymerization, some oligomers can reattach to the surface, becoming part of the particles again. Since the oligomers quickly enter the solution, they do not form a layer that hinders EG diffusion. Therefore, the above reaction process conforms to the characteristics of a shrinking core model without an ash layer (Yoshioka

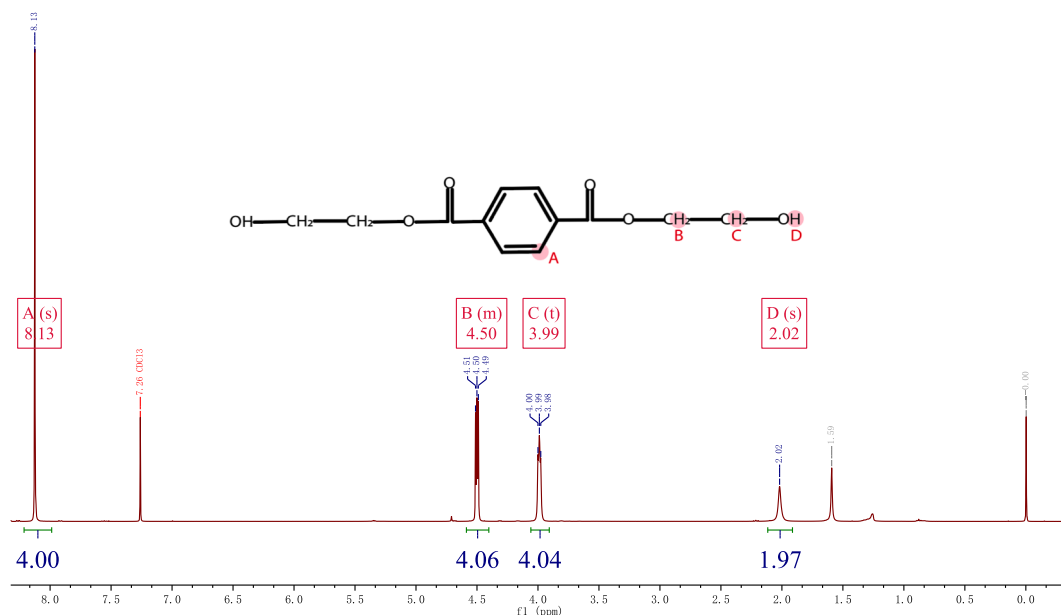
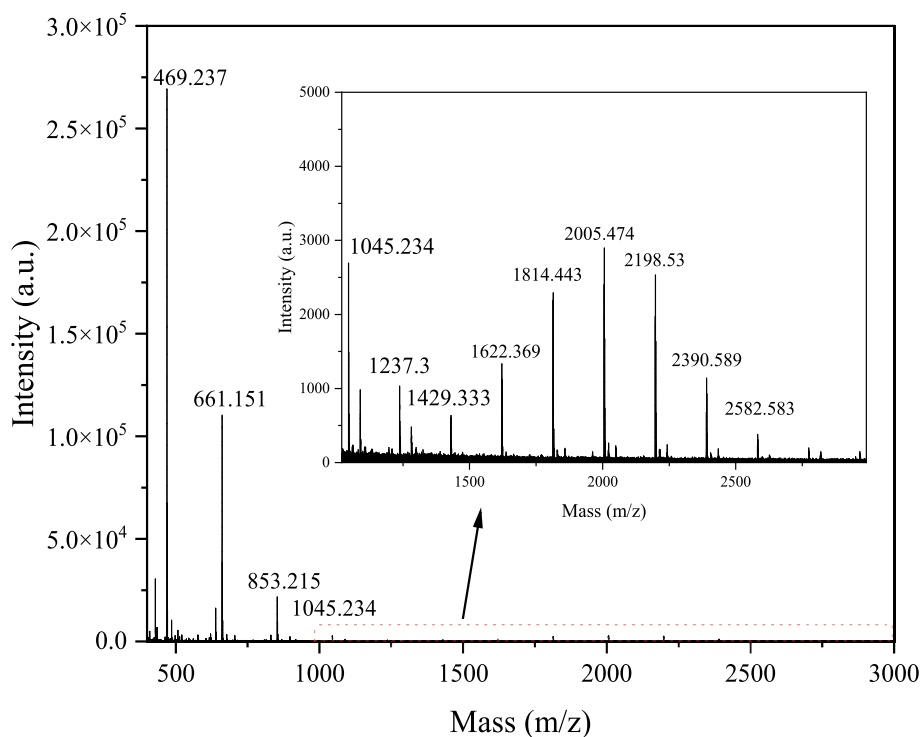
Fig. 3. ^1H NMR spectrum of BHET product.

Fig. 4. MALDI TOF-MS spectrum of oligomers.

et al., 2001; Yoshioka et al., 1998). Based on a similar catalytic reaction mechanism, the oligomers in the solution react with EG to convert into BHET. A reverse reaction also exists, leading to an equilibrium between oligomers and BHET.

The repeat units of PET molecules, consisting of terephthalic acid and EG linked through ester bonds, remain conserved during the glycolysis process. These repeat units are redistributed among unreacted PET particles, oligomers, and BHET and were tracked throughout glycolysis to facilitate the integration of the heterogeneous and homogeneous models.

The initial moles of PET repeat units (n_T) can be expressed as

$$n_T = m_{\text{PET},\text{ini}}/M_{\text{PET}} = \frac{4}{3}\pi R_s^3 T \rho_{\text{PET}}/M_{\text{PET}} \quad (3)$$

where M_{PET} is the molar mass of the repeat unit. R_s , ρ_{PET} , and T are the diameter, density, and number of PET particles, respectively.

The excess molar ratio of EG relative to the repeat units in PET (S) can be calculated by

$$S = n_{\text{EG},\text{ini}}/n_T \quad (4)$$

where $n_{\text{EG},\text{ini}}$ is the initial number of moles of EG.

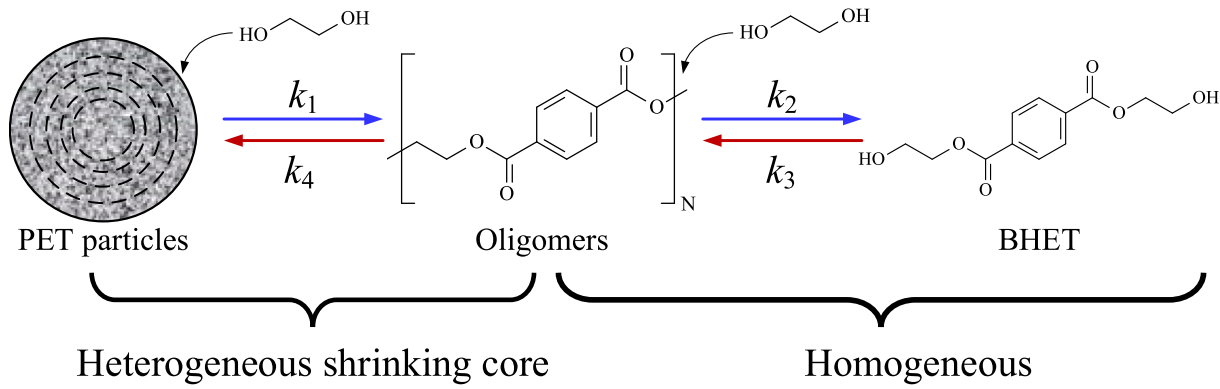


Fig. 5. Proposed reaction process of PET glycolysis.

Define n_{PET} as the moles of PET repeat units within unreacted PET particles. Both the depolymerization rate of oligomers from the particle surface and the repolymerization rate of oligomers to the particle surface are evidently related to the specific surface area of the particles. It is assumed that the relationship between the depolymerization rate and the EG concentration (C_{EG}) and the relationship between the repolymerization rate and the oligomer concentration ($C_{Oligomer}$) are consistent with the first-order reaction characteristics. The change in n_{PET} over reaction time according to the shrinking core model can be expressed by

$$\frac{dn_{PET}}{dt} = -Tk_1C_{EG}4\pi R_c^2 + Tk_4C_{Oligomer}N4\pi R_c^2 \quad (5)$$

where R_c is the diameter of the unreacted PET particles.

Define $n_{Oligomer}$ as the number of moles of oligomers with a distribution of polymerization degree. For simplification, the oligomers are considered as a lumped component with an average degree of polymerization N . Assume that the relationship between the conversion rate of oligomers to BHET and the concentrations of oligomers ($C_{Oligomer}$) and EG (C_{EG}) is consistent with the first-order reaction characteristics. At the same time, the relationship between the conversion rate of BHET to oligomers and the concentration of BHET is consistent with the second-order reaction characteristics. Therefore, the change of $n_{Oligomer}$ in the solution with reaction time can be expressed by

$$\frac{1}{V_L} \frac{dn_{Oligomer}}{dt} = k_1C_{EG}4\pi R_c^2 \frac{T}{NV_L} - k_2C_{EG}C_{Oligomer} + k_3C_{BHET}^2 - k_4C_{Oligomer}4\pi R_c^2 \frac{T}{V_L} \quad (6)$$

where V_L is the volume of the liquid phase.

Y_{PET} , $Y_{Oligomer}$ and Y_{BHET} are further defined as the mole fractions of PET repeat units occupied by unreacted PET particles, oligomers and BHET, respectively.

$$Y_{PET} = \frac{n_{PET}}{n_T} \quad (7)$$

$$Y_{Oligomer} = \frac{Nn_{Oligomer}}{n_T} \quad (8)$$

Y_{PET} is numerically equal to $(100 - X_{PET})/100$, while Y_{BHET} is numerically equal to $P_{BHET}/100$. According to the characteristics of spherical particles, $R_c = R_s Y_{PET}^{1/3}$ can be obtained. Combining Eqs. (5) and (6), the following equation can be derived.

$$\frac{dY_{PET}}{dt} = -\frac{Tk_1C_{EG}4\pi R_s^2 Y_{PET}^{\frac{2}{3}}}{n_T} + Tk_4Y_{Oligomer}4\pi R_s^2 Y_{PET}^{\frac{2}{3}}/V_L \quad (9)$$

$$\frac{dY_{Oligomer}}{dt} = \frac{k_1C_{EG}4\pi R_s^2 Y_{PET}^{\frac{2}{3}}T}{n_T} - k_2C_{EG}Y_{Oligomer} + \frac{Nk_3Y_{BHET}^2n_T}{V_L} - Tk_44\pi R_s^2 Y_{PET}^{\frac{2}{3}}Y_{Oligomer}/V_L \quad (10)$$

Based on the molar volumes of the substances entering and leaving the liquid phase, the volume of the liquid phase can be derived as

$$V_L = n_T[SV_{EG} - \left(Y_{BHET} + \frac{Y_{Oligomer}}{N}\right)V_{EG} + Y_{BHET}V_{BHET} + \frac{Y_{Oligomer}}{N}V_{Oligomer}] \quad (11)$$

where V_{EG} , V_{BHET} and $V_{Oligomer}$ are the calculated molar volumes of EG, BHET and oligomers.

The value of C_{EG} can be derived as

$$C_{EG} = \frac{N_T}{V_L} (S - Y_{BHET} - Y_{Oligomer}/N) \quad (12)$$

3.3.2. Parameter estimation for reaction kinetic model

Combining Equations (9) through (12), a complete reaction kinetic model for PET glycolysis is developed. The parameters required for estimation are the average degree of polymerization N and the rate constants k_1 to k_4 . These parameters were estimated using the nonlinear least squares method as described in Eq. (13) (Chen et al., 2021; Liu et al., 2016).

$$\begin{cases} RSS(N, k_{1-4}) = \sum_{i=1}^n \sum_{j=1}^m (Y_{ij,predicted} - Y_{ij,measured})^2 \\ N, k_{1-4} \geq 0 \end{cases} \quad (13)$$

where $Y_{ij,predicted}$ and $Y_{ij,measured}$ represent the model predicted mole fraction and the experimentally measured mole fraction of the j^{th} data set in the i^{th} series of experiments, respectively. RSS is the residual sum of squares between $Y_{ij,predicted}$ and $Y_{ij,measured}$.

The lsqnonlin function provided by Matlab R2024 software package was used to estimate N and rate constants k_1 to k_4 . The correlation between the model predicted distribution and the experimentally measured distribution was checked based on residual error analysis and F-test with a significance level of 5%. The calculation results show that the RSS value reaches the minimum when N is 2, which is consistent with the MALDI TOF-MS characterization on the oligomers. Table 1 lists

Table 1
Estimated reaction rate constants at different temperatures.

Temperature (°C)	k_1 (m/min)	k_2 (m ³ /mol.min)	k_3 (m ³ /mol.min)	k_4 (m/min)
140	4.79E-08	5.56E-06	1.48E-05	4.74E-06
160	5.03E-07	2.54E-05	4.84E-05	3.68E-05
180	4.06E-06	1.01E-04	1.43E-04	2.25E-04

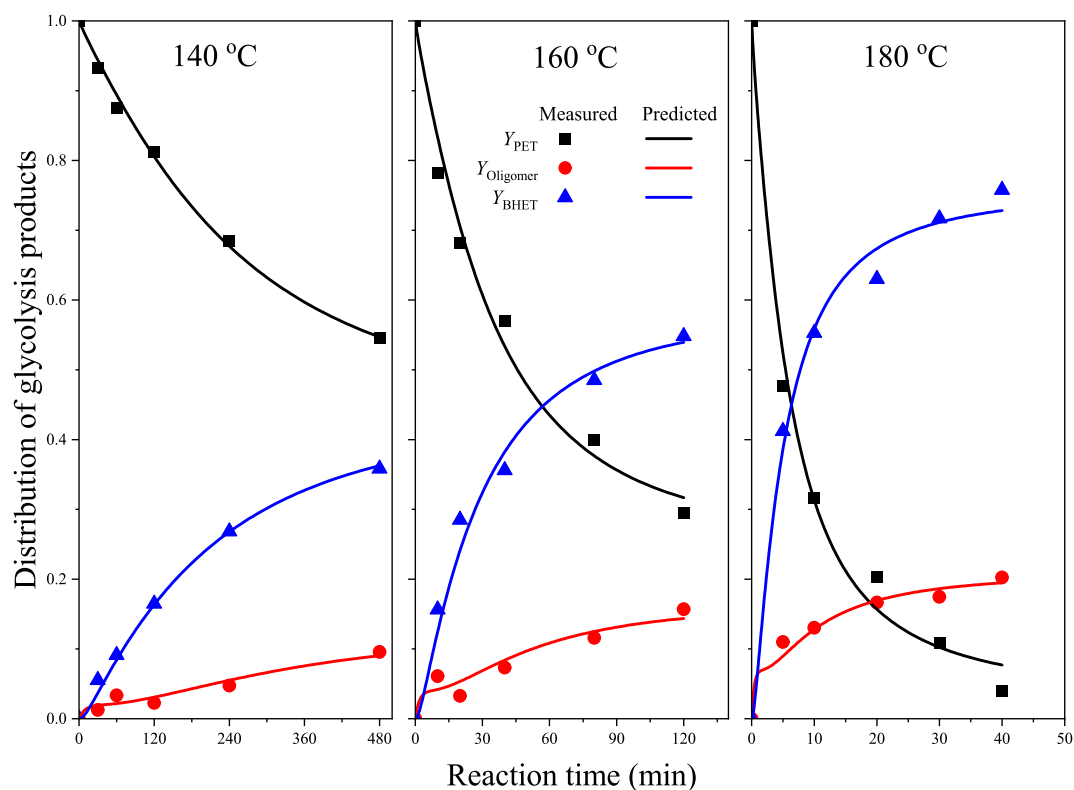


Fig. 6. Comparison between measured and predicted product distribution of PET glycolysis.

the calculated reaction rate constants at different temperatures, while Fig. 6 shows the comparison between the experimental measured values and the model predicted values of Y_{PET} , $Y_{Oligomer}$ and Y_{BHET} .

At relatively low temperatures of 140 and 160 °C, the model predictions almost coincide with the experimental values. According to the model prediction, at the beginning of the reaction, the concentration of

oligomers in the solution may be slightly higher than the corresponding value of BHET. However, as the reaction time increases, oligomers are rapidly converted to BHET, causing the concentration of BHET to exceed that of oligomers. When the temperature is increased to 180 °C, the model predictions still reproduce the overall trend of the experimental values with high accuracy.

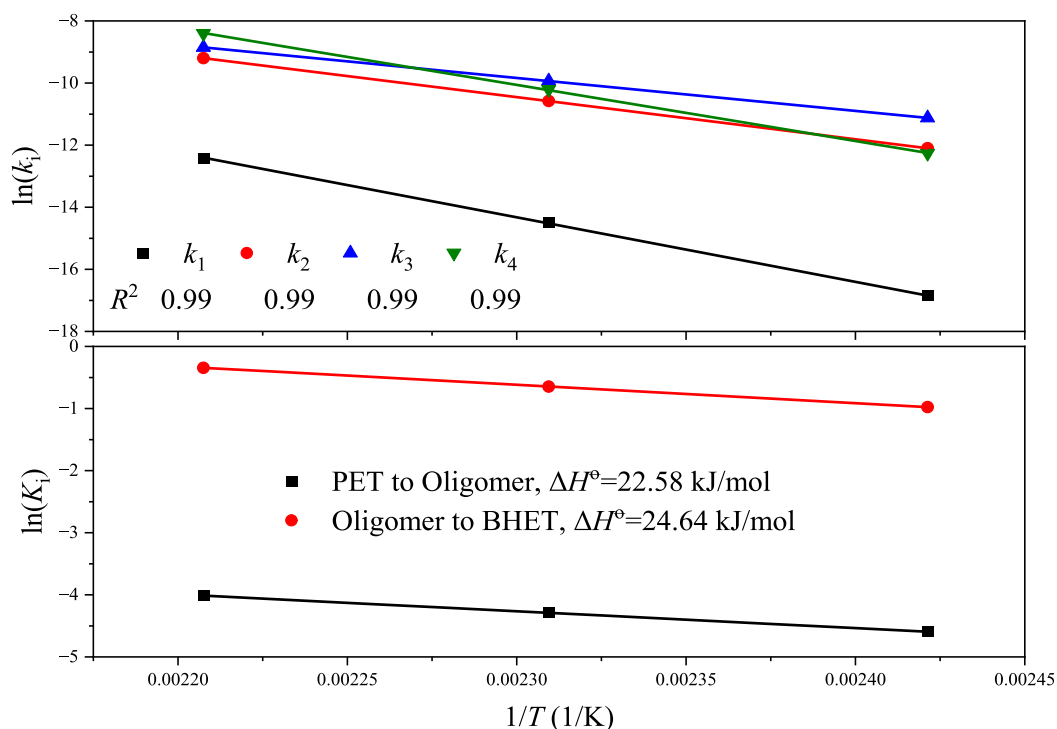


Fig. 7. Arrhenius plot of $\ln k$ vs. $1/T$ (up) and Van 't Hoff plot of $\ln K$ vs. $1/T$ (down).

According to the $\ln k$ vs. $1/T$ relationship shown in Fig. 7, the activation energy and pre-exponential factor of the reactions involved in glycolysis were further calculated based on the Arrhenius formula, and the results are listed in Table 2.

During the linear regression of reaction kinetic parameters shown in Fig. 7, the coefficient of determination (R^2) reached 0.99, indicating that the model can predict the glycolysis trends with high precision. The activation energies for the depolymerization of PET to oligomers and the depolymerization of oligomers to BHET are higher than the reverse reactions. That is to say, both the heterogeneous shrinking core reaction step and the homogeneous reaction step require overcoming higher reaction barriers for the forward reactions to proceed effectively. Given that the conversion of PET to oligomers has the highest activation energy of 172.71 kJ/mol, it can be inferred that the depolymerization of PET to oligomers is the rate-determining step in the entire glycolysis process. Increasing the reaction temperature can significantly accelerate glycolysis by overcoming these activation energy barriers more efficiently.

The standard enthalpy changes ($\Delta H^\ominus$) for the heterogeneous shrinking core and homogeneous reaction steps were calculated using the Van't Hoff equation, with the $\ln k$ vs. $1/T$ relationship shown in Fig. 7. The calculated values of $\Delta H^\ominus$ for the two steps are 22.58 and 24.64 kJ/mol, respectively, consistent with the endothermic characteristics suggested by López-Fonseca (López-Fonseca et al., 2011). From the relationship between thermodynamic parameters, the calculated standard entropy change for the shrinking core depolymerization is 16.46 J/mol·K. Considering the endothermic nature of the reaction, PET depolymerization can be regarded as an entropy-driven process at high temperatures (Ruvolo-Filho and Curti, 2006). From a thermodynamic perspective, the involvement of high temperatures shifts the reaction to become spontaneous. From a reaction kinetic standpoint, increasing the temperature accelerates the conversion of both PET to oligomers and oligomers to BHET.

• Validation of reaction kinetic model

During the development of the reaction kinetic model for PET glycolysis, certain simplifications and assumptions were introduced, necessitating the verification of the applicability of the model. Glycolysis was performed using 0.15 mm PET particles at 200 to 240 °C and 0.85 mm PET particles at 140 and 180 °C. Simultaneously, the corresponding product distributions were predicted based on the reaction kinetic parameters listed in Table 2.

Experimental data obtained at temperatures of 200 °C and above are not suitable for reaction kinetics regression due to the interference of the preheating process. PET particles with a diameter of 0.15 mm are nearly completely converted within 5 min of reaction at 200 °C, and further extension of the reaction time does not result in significant changes in the product distribution. However, the product distribution obtained at high temperatures remains valuable for evaluating the ability of the model to predict reaction equilibrium. Fig. 8 shows a comparison of predicted and measured equilibrium BHET fractions in PET glycolysis products at high temperatures. According to model predictions, the equilibrium values of Y_{BHET} at 200, 220, and 240 °C are 0.81, 0.85, and 0.86, respectively. The maximum deviation between the experimentally measured equilibrium values and the predicted values does not exceed 1.5 %, confirming the high accuracy of the developed kinetic model in

predicting the reaction equilibrium of PET glycolysis at elevated temperatures.

Fig. 9 shows a comparison between the predicted and experimentally measured glycolysis product distributions for large PET particles of 0.85 mm. An increase in particle size leads to a substantial reduction in the specific surface area available for contact with the liquid phase. Compared with the data in Fig. 2, the increase in PET particle size hinders the conversion to oligomers and BHET, further supporting the necessity of incorporating the shrinking core model into the glycolysis kinetic framework. Although the model parameters listed in Table 2 were estimated based on experimental data for 0.15 mm PET particles, the kinetic model effectively predicted the product distribution for 0.85 mm PET particles. At 140 °C, the predicted product distribution closely matched the experimental results. Even at 180 °C, where deviations between model predictions and experimental values were more pronounced, the model still accurately captured the overall trend in product distribution. This further validates the generalizability of the developed reaction kinetic model.

3.4. PET glycolysis in batch or CSTR mode

The purpose of reaction kinetics research is to provide a basis for process optimization and reactor design. Therefore, the reaction kinetic model developed in this study is applied to engineering analysis. PET glycolysis can be performed using either batch processes or continuous stirred tank reactors CSTR in series. Regardless of the operational method chosen, the reaction temperature is a critical process parameter that must be determined first. The reaction temperature not only dictates the reaction kinetic and thermodynamic characteristics of the process but also significantly impacts the operational method and the dimensions of the equipment required.

Fig. 10 shows the effect of reaction temperature on PET conversion and BHET yield during PET glycolysis in batch mode. At 140 °C, the reaction approaches equilibrium only when the reaction time is extended to 24 h or longer. At equilibrium, the PET conversion is just 54 %, and the BHET yield is 42 %. Increasing the reaction temperature significantly accelerates PET glycolysis and improves BHET yield. When the reaction temperature is raised to 200 °C, the time required to achieve complete PET conversion is reduced to approximately 7 min. Concurrently, the BHET yield increases to 82 %. Further increasing the reaction temperature to 220 °C or higher accelerates PET conversion even more, but the improvement in BHET yield is limited. At 220 and 240 °C, the equilibrium BHET yields are 85 % and 86 %, respectively.

Conducting PET glycolysis using a series of CSTRs enables the continuous process and integrated energy utilization. However, back-mixing leads to an increase in product concentration, which significantly reduces the glycolysis rate. Taking the example of a glycolysis process with a PET processing rate of 100 kg/h, an EG/PET mass ratio of 4:1, PET particle diameter of 0.15 mm, and a total liquid volume of 0.01 m³ in the reactors, Fig. 11 shows the PET conversion and BHET yield achievable within the temperature range of 200 to 240 °C under equal volume multi-reactor series conditions.

When the reaction temperature is set at 200 °C, a single CSTR achieves a PET conversion of 53.8 % and a BHET yield of 42.7 %. The use of multiple CSTRs in series effectively mitigates the adverse impact of back-mixing on the reaction rate. As the number of CSTRs in series increases, both the PET conversion and BHET yield continue to rise. However, even with 10 CSTRs in series, the maximum PET conversion and BHET yield achieved are 86 % and 65 %, respectively. Increasing the reaction temperature significantly accelerates glycolysis. At 220 °C, complete PET degradation requires three CSTRs in series. The PET conversions at the outlet of each CSTR are 70.5 %, 95.2 %, and 99.9 %, respectively, while the BHET yields are 55.7 %, 78.3 %, and 83.8 %. Due to the decreasing concentration driving force, the incremental PET conversion achieved by each CSTR shows a marked decline. For a three-CSTR configuration, the BHET yield has already approached equilibrium

Table 2

Arrhenius parameters of reactions involved in PET glycolysis.

	PET → Oligomer	Oligomer → BHET	BHET → Oligomer	Oligomer → PET
$\ln(A)$	33.45	20.78	14.58	31.47
E_a (kJ/ mol)	172.71	112.90	88.26	150.13

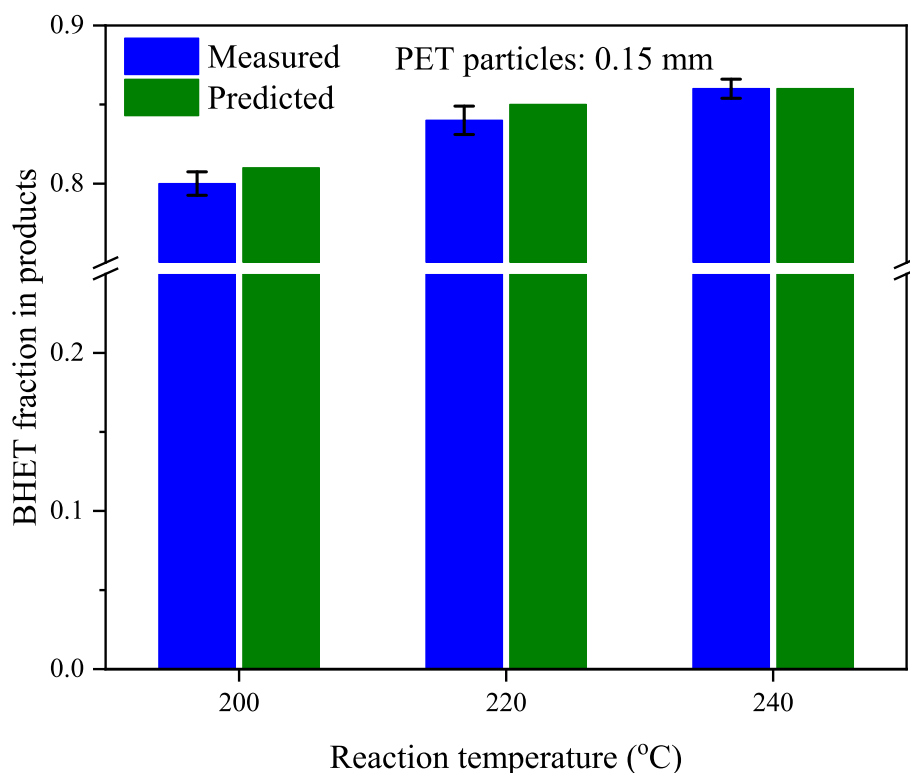


Fig. 8. Comparison of predicted and measured equilibrium BHET fractions in PET glycolysis products at high temperatures.

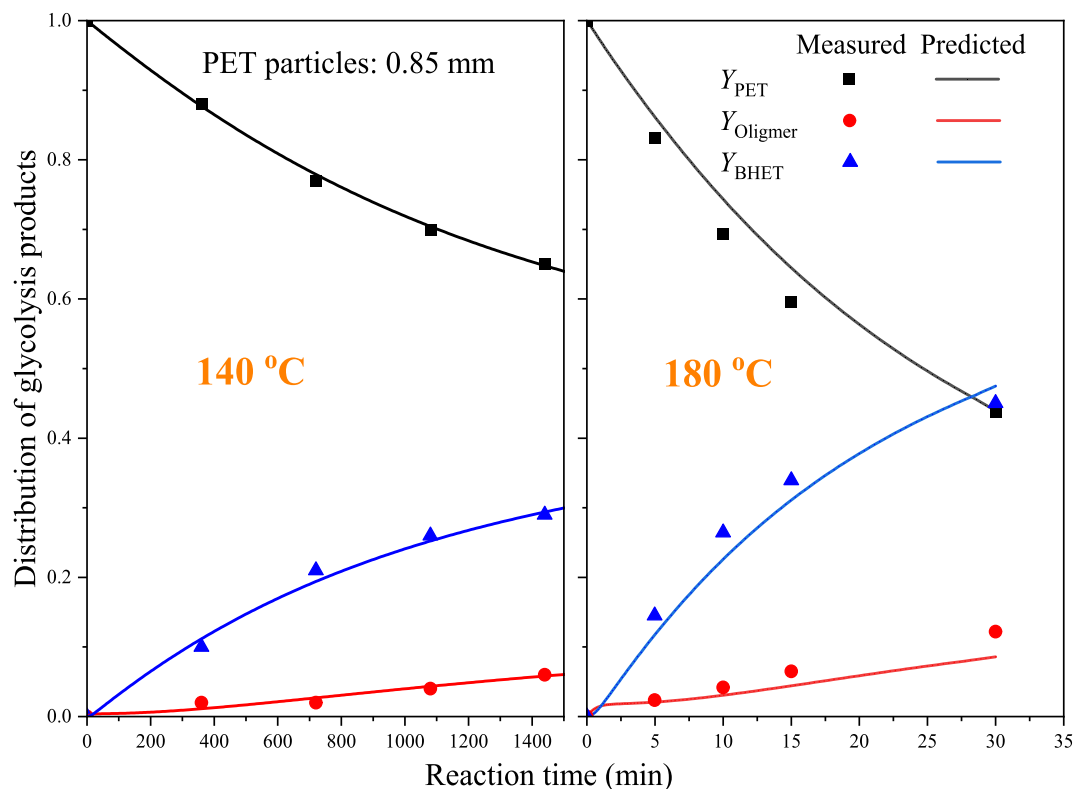


Fig. 9. Comparison of predicted and measured values of glycolysis product distribution of large particle PET.

value. To reduce control complexity and equipment investment, industrial operations typically aim to minimize the number of CSTRs in series. Therefore, further increasing the glycolysis reaction temperature is necessary. At 240 °C, only one CSTR is required to achieve a 99.2 % PET

conversion. At this temperature, the BHET yield reaches 85 %, which is close to the equilibrium yield. Although a single CSTR reactor at 240 °C can nearly complete the conversion of the specified PET processing capacity, the reactor volume is approximately four times larger than that

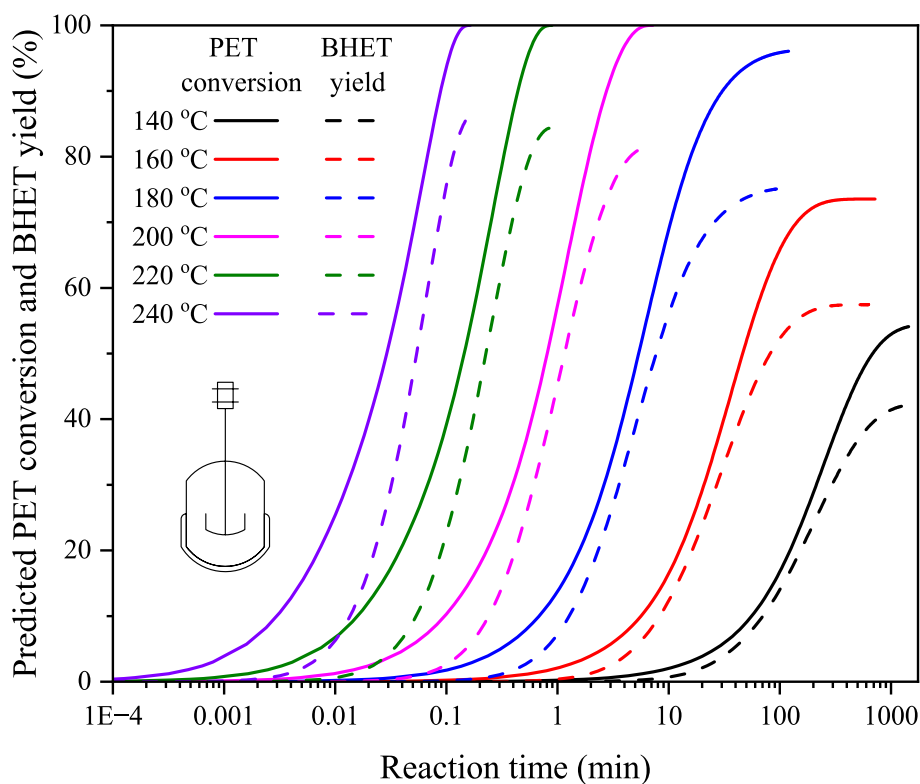


Fig. 10. Predicted glycolysis performance at different temperatures in batch mode.

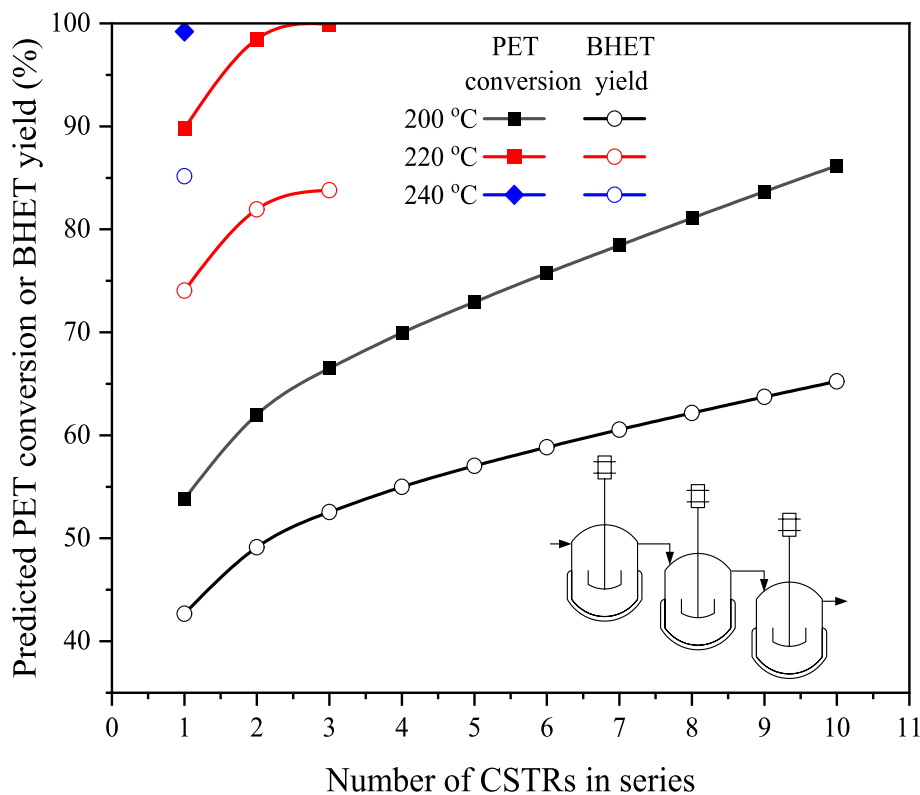


Fig. 11. Predicted glycolysis performance at different temperatures in series CSTR mode.

of a plug flow reactor.

The first step of PET glycolysis involves a heterogeneous shrinking core reaction, with efficiency directly related to the specific surface area of the particles. Fig. 12 shows the reaction time required for PET

particles of different sizes to achieve 99.5 % conversion at temperatures ranging from 200 to 240 °C. At the relatively low temperature of 200 °C, an increase in particle size significantly extends the reaction time. For instance, PET particles with a diameter of 0.15 mm require 4.3 min for

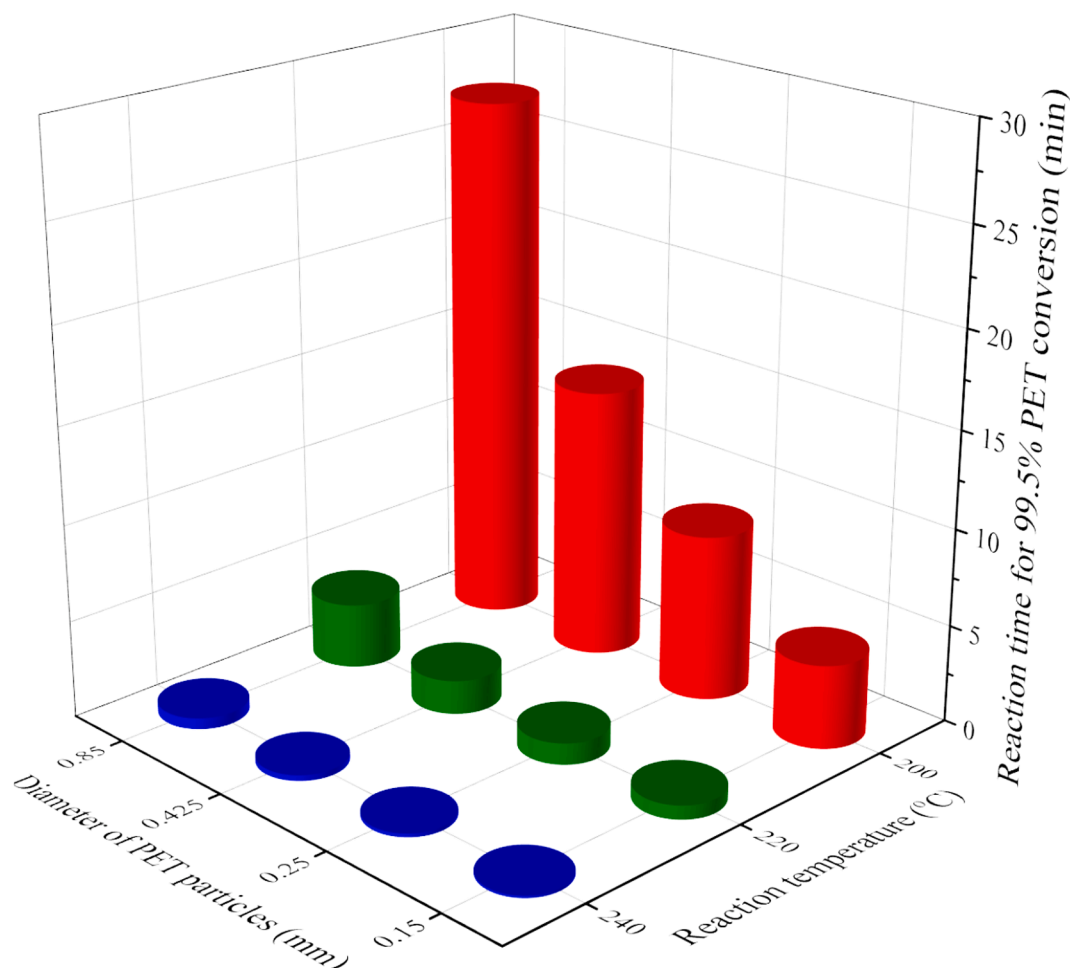


Fig. 12. Effect of PET particle diameter on reaction time for PET conversion to 99.5%.

near-complete conversion, whereas particles with a diameter of 0.85 mm require 27.3 min. Since the depolymerization of PET to oligomers has the highest activation energy in the entire reaction network, elevating the reaction temperature mitigates the sensitivity of glycolysis efficiency to particle size. At 240 °C, the time required for near-complete PET conversion only increases marginally from 0.14 min for 0.15 mm particles to 0.58 min for 0.85 mm particles.

For batch-mode glycolysis, increasing the size of PET particles does not pose significant challenges to industrial operations. Extending the reaction time can adequately compensate for the decrease in reaction rate caused by the larger particle diameter. However, for CSTR operations, particle size directly affects the CSTR volume and the number of CSTRs needed in series, especially at relatively lower reaction temperatures. Referring to the calculations shown in Fig. 11 for the same PET processing rate, when the PET particle diameter is 0.85 mm, the required volume of a single CSTR reactor to achieve 99.5 % PET conversion at reaction temperatures of 200, 220, and 240 °C are 5, 0.5, and 0.07 m³, respectively.

4. Conclusions

The glycolysis of PET in EG involves a combination of heterogeneous and homogeneous reactions. The reaction products consist of unreacted PET particles, oligomers with a certain degree of polymerization, and BHET, with an equilibrium constraint between oligomers and BHET. Liquid-solid mass transfer, surface reactions, and catalyst dosage all influence the reaction rate. Preliminary experiments determined the optimal operating conditions and raw material ratios: a stirring rate of

200 rpm, an EG/PET mass ratio of 4:1, and a catalyst dosage of 1 wt%. The subsequent reaction kinetic experiments confirmed that reaction temperature has a decisive influence on the glycolysis rate and reaction equilibrium. Characterization of the oligomers indicated a distribution of degrees of polymerization, predominantly dimers.

By tracking the distribution of PET repeat units in the products and combining the heterogeneous shrinking core reaction model with the homogeneous reaction model, a kinetic model for PET glycolysis was developed. This model accurately replicates the data from reaction kinetic experiments and predicts the equilibrium conversions at higher temperatures, as well as the reaction behavior of larger PET particles. The step involving the depolymerization of PET to oligomers has the highest activation energy of 172.71 kJ/mol, making it the rate-determining step in the entire reaction network. Additionally, the standard reaction enthalpy and entropy change for this step are 22.58 kJ/mol and 16.46 J/mol·K, respectively, indicating that the glycolysis process is an entropy-driven process requiring high temperatures for effective conversion.

Increasing the reaction temperature effectively reduces the time required for complete conversion of PET of a given particle size, enhances BHET yield, and decreases the sensitivity of the conversion time to PET particle size. At 240 °C, approximately 86 % BHET yield can be achieved. Although using the CSTR operation mode allows for continuous PET degradation, back-mixing negatively impacts the reaction rate. For glycolysis operating with multiple CSTRs in series, it is recommended to use high temperatures and small PET particles to reduce the volume of individual reactors and the number of reactors in series.

CRediT authorship contribution statement

Ling-Han Wang: Writing – original draft, Investigation. **Sen Lan:** Investigation. **Zi-Bin Huang:** Methodology. **Jing-Yi Yang:** Software. **Jian-Hong Gong:** Supervision. **Pei-Qing Yuan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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